



CATHOLIC SECONDARY SCHOOLS
ASSOCIATION OF NSW

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Centre Number

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Student Number

DO NOT REMOVE PAPER FROM EXAM ROOM

2019

TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

Physics

Morning Session
Tuesday, 13 August 2019

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- Use Multiple-Choice Answer Sheet provided
- NESA-approved calculators may be used
- A data sheet and formulae sheets are provided SEPARATELY. Ensure you use the replacement Periodic Table sheet
- Write your Centre Number and Student Number on the top of this page

Total marks - 100

Section I

Pages 2-10

20 marks

- Attempt Questions 1-20
- Allow about 35 minutes for this section

Section II

Pages 11-26

80 marks

- Attempt Questions 21-33
- Allow about 2 hours and 25 minutes for this section

Disclaimer

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Section I

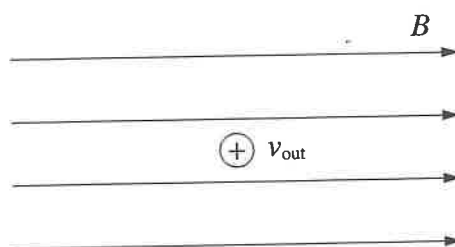
20 marks

Attempt Questions 1-20

Allow about 35 minutes for this part

Use the Multiple-Choice Answer Sheet for Questions 1-20.

- 1 A positively charged particle is moving through a magnetic field with a velocity v .



If the charge is moving out of the page, what is the direction of the force F on the particle?

- (A) to the right
 - (B) to the left
 - (C) downwards
 - (D) upwards
- 2 Diffraction is the name given to the:
- (A) change of direction when waves cross the boundary between one medium and another.
 - (B) addition of two coherent waves to produce a stationary wave pattern.
 - (C) bending of waves round an obstacle.
 - (D) splitting of white light into colours.

- 3 Hertzprung-Russell diagrams are used to classify stars using a variety of different characteristics.

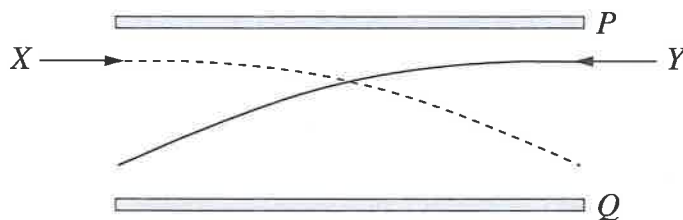
Which of the following is NOT a characteristic that can be determined using a Hertzprung-Russell diagram?

- (A) surface temperature
- (B) apparent brightness
- (C) spectral class
- (D) luminosity

- 4 What is the angular velocity of the Earth in its orbit around the Sun?

- (A) $2.0 \times 10^{-7} \text{ rad s}^{-1}$
- (B) $7.2 \times 10^{-5} \text{ rad s}^{-1}$
- (C) $1.1 \times 10^{-5} \text{ rad s}^{-1}$
- (D) $4.1 \times 10^{-3} \text{ rad s}^{-1}$

- 5 The diagram shows the paths of two charged particles, X and Y , during their passage between a pair of oppositely charged metal plates, P and Q .



The plates are charged such that the electric field between them is directed from Q to P .

Which charges on X and Y will produce the observed paths?

	X	Y
(A)	–	–
(B)	–	+
(C)	+	–
(D)	+	+

- 6 James Clerk Maxwell's revolutionary theory of electromagnetism was first proposed in 1865, it was not well accepted by the scientific community until some 20 years later.

What was the reason for this?

- (A) There was little experimental evidence available at the time to support the theory.
- (B) The photoelectric effect had shown that light had particle-like properties.
- (C) The speed of light had not been measured experimentally.
- (D) Spectroscopy had not yet been developed.

- 7 If a wave can be polarised, it must be:

- (A) an electromagnetic wave.
- (B) a longitudinal wave.
- (C) a mechanical wave.
- (D) a transverse wave.

- 8 In 1909, Robert Millikan and Harvey Fletcher conducted the oil drop experiment to determine the:

- (A) charge to mass ratio on an electron.
- (B) nature of cathode rays.
- (C) charge on an electron.
- (D) charge on a proton.

- 9 A satellite P orbits the Earth at a distance r between the two centres with a period T_P . Another satellite Q orbits at a distance of $6r$ in a time T_Q . The ratio of T_P to T_Q is:

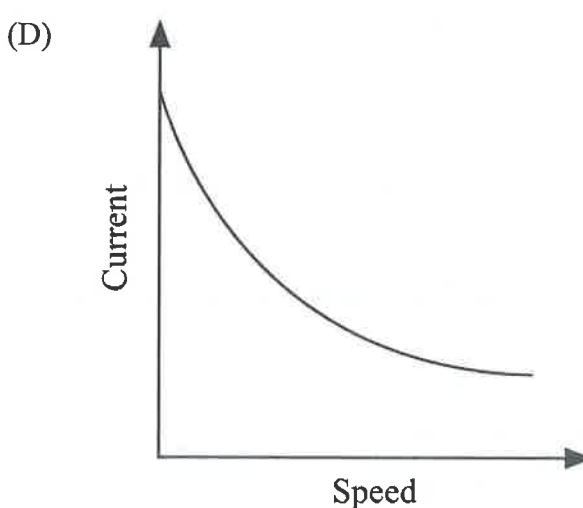
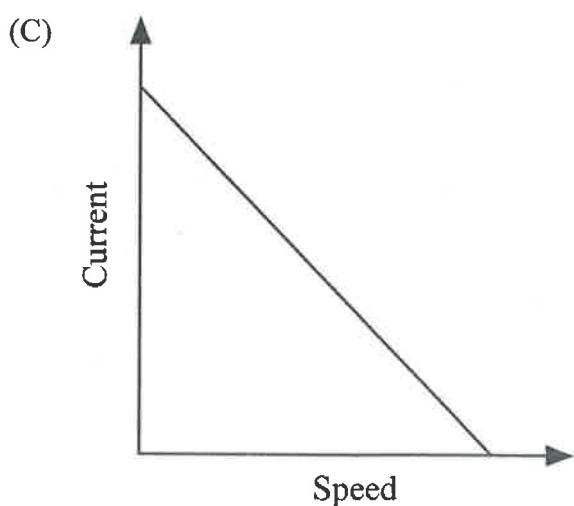
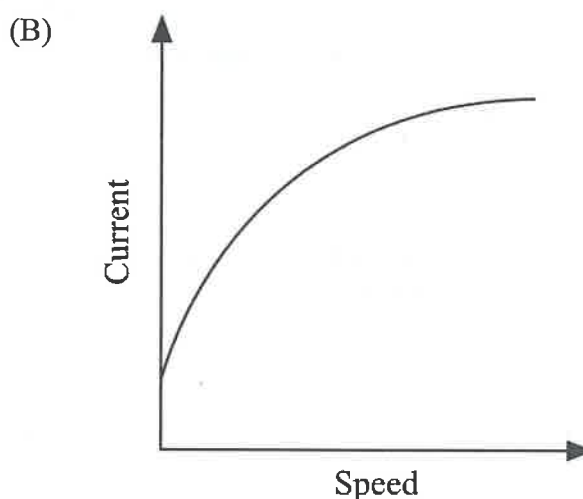
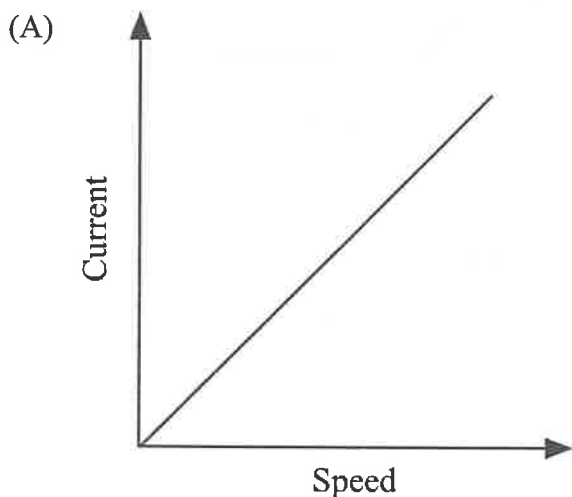
- (A) 1:15
- (B) 1:6
- (C) 6:1
- (D) 15:1

10 Which of the following is NOT a function of transformers in the large-scale distribution of electrical energy?

- (A) Transformers increase voltage across transmission lines for efficient power distribution.
- (B) Transformers reduce energy losses in transmission lines by decreasing current.
- (C) Transformers reduce energy losses in transmission lines by increasing current.
- (D) Transformers decrease voltage across household circuits for safety reasons.

11 A DC electric motor is connected to a power supply operating at a constant voltage. The speed of the motor is varied by applying a friction brake to the axle.

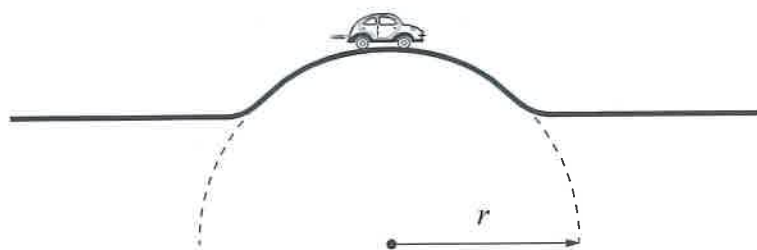
Which graph correctly shows how the current through the motor windings varies with speed?



12 Which fundamental quantity required that its unit of measurement be redefined following acceptance of the theory of special relativity?

- (A) luminous intensity
- (B) length
- (C) mass
- (D) time

13 A car and driver with mass m moves at a speed v over a bridge which is the arc of a circle of radius r .



Which of these expressions represents the maximum speed of the car so that it remains in contact with the road?

- (A) rg
- (B) $\sqrt{rg}$
- (C) mrg
- (D) $\sqrt{mrg}$

14 An electron in a hydrogen atom transitions from a quantum state with number $n = 3$ to the ground state.

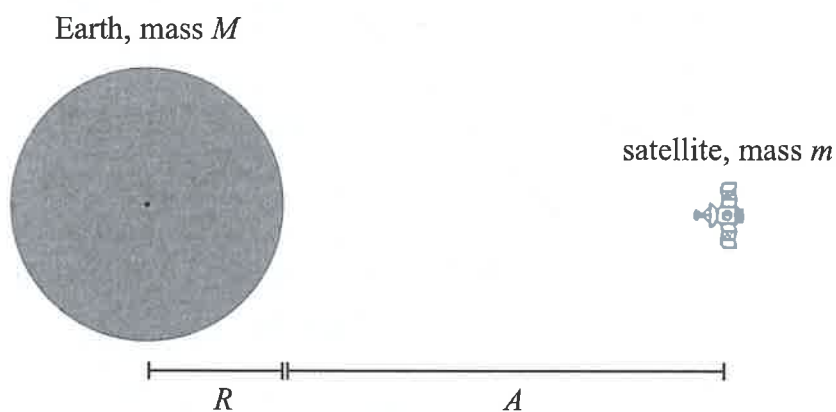
The wavelength of the emitted photon is closest to:

- (A) 95 nm
- (B) 100 nm
- (C) 105 nm
- (D) 110 nm

15 Which particles are the only constituents of hadrons?

- (A) neutrinos
- (B) electrons
- (C) quarks
- (D) bosons

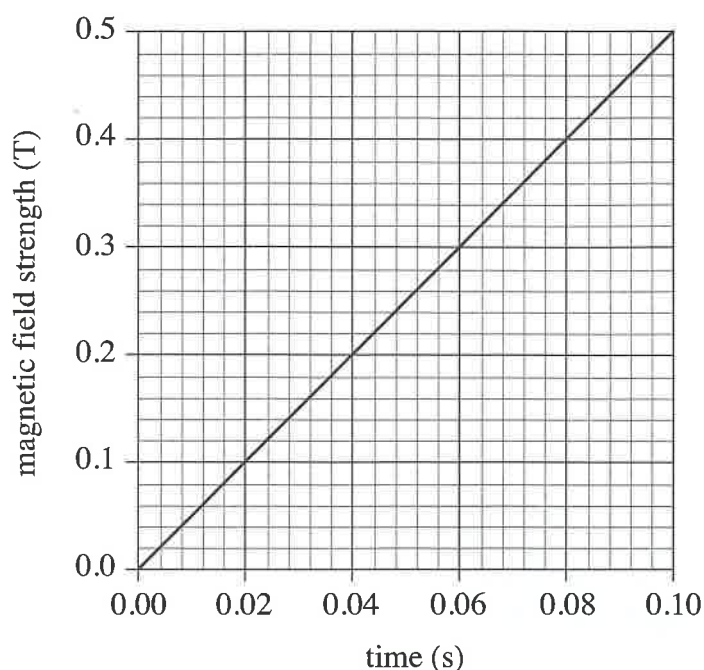
16 A satellite of mass m orbits the Earth at an altitude A with an orbital velocity v above the surface of the Earth as shown. The Earth has a mass M and a radius R .



Which expression represents the total energy of the satellite?

- (A) $-\frac{GMm}{2(R+A)}$
- (B) $-\frac{GM}{2(R+A)}$
- (C) $-\frac{GMm}{(R+A)}$
- (D) $-\frac{GMm}{R}$

- 17 A circular coil of 100 turns with a radius of 2.0 cm is placed in a changing magnetic field. The angle between the magnetic field lines and the plane of the coil is 60° . The graph below shows the variation with time of the magnetic field strength.



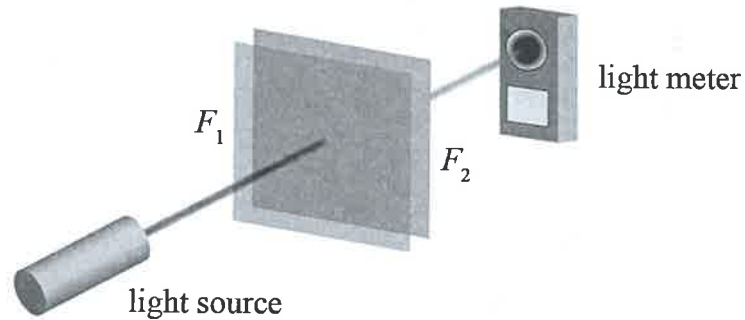
What is the magnitude of the *emf* induced in the coil?

- (A) -0.63 V
 - (B) -0.54 V
 - (C) -0.31 V
 - (D) -0.16 V
- 18 A parallel beam of white light passes through a diffraction grating. Orange light of wavelength 600 nm in the fourth order diffraction maximum coincides with blue light in the fifth order diffraction maximum.

What is the wavelength of the blue light?

- (A) 450 nm
- (B) 480 nm
- (C) 500 nm
- (D) 750 nm

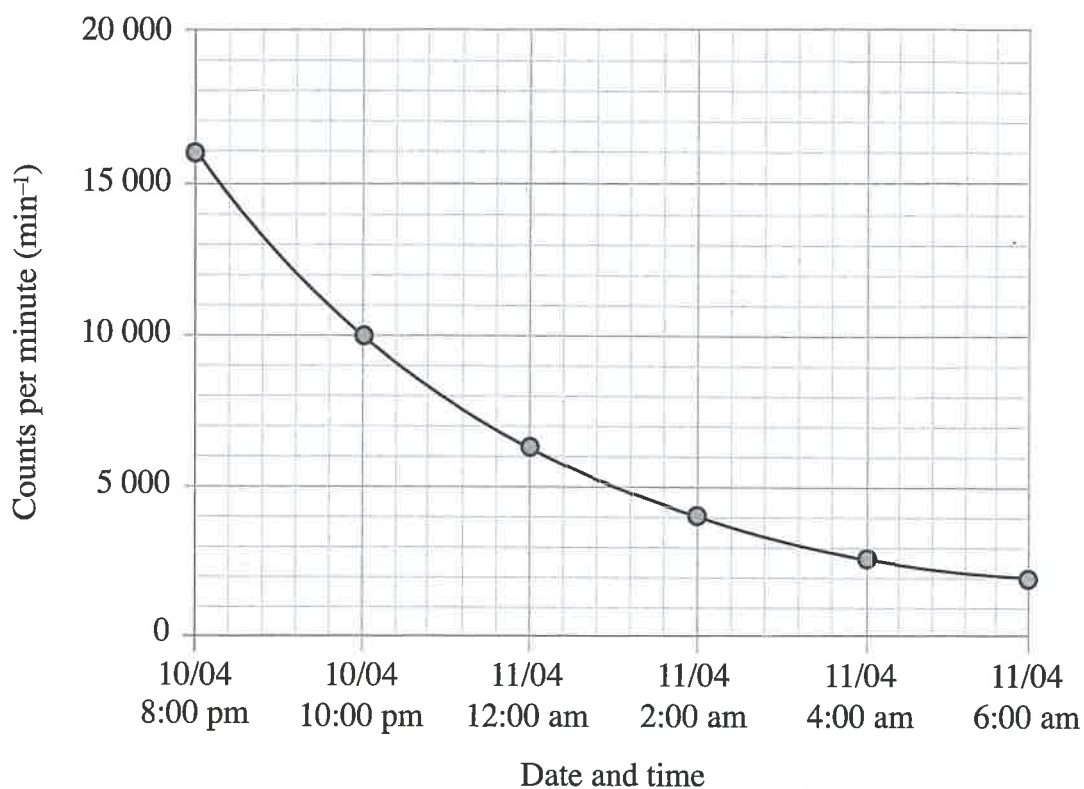
- 19 In an experiment to verify Malus' law, two students measured the intensity of monochromatic light passing through two polarising filters. One student rotated the second filter F_2 with respect to the axis of the first filter F_1 . The other student held a light meter in front of F_2 to measure the intensity of the light that had passed through both filters.



Which of the following would most significantly improve both the accuracy and reliability of the experiment?

- (A) taking many readings of light intensity for each angle of rotation
- (B) using white light instead of monochromatic light
- (C) decreasing the intensity of the incident light
- (D) securing the light detector to the benchtop

- 20 The graph below shows the decay rate for a radioisotope in the blood of a patient after a nuclear medicine procedure.



What is the decay constant λ for technetium-99m?

- (A) $9.6 \times 10^{-5} \text{ s}^{-1}$
- (B) $6.4 \times 10^{-5} \text{ s}^{-1}$
- (C) $4.8 \times 10^{-5} \text{ s}^{-1}$
- (D) $3.2 \times 10^{-5} \text{ s}^{-1}$

Section II

80 marks

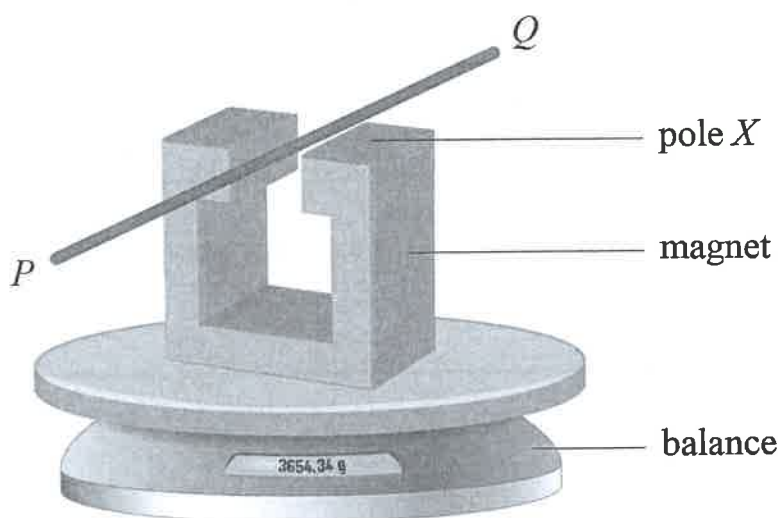
Attempt Questions 21-33

Allow about 2 hours and 25 minutes for this section

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
 - Show all relevant working in questions involving calculations.
 - Extra writing space is provided on page 27. If you use this space, clearly indicate which question you are answering.
-

Question 21 (5 marks)

A large horseshoe magnet produces a uniform magnetic field B between its poles. The magnet is placed on a digital balance and a secured copper rod placed between its poles as shown. The rod PQ is positioned perpendicularly to the magnetic field and horizontally to the pan of the balance. The length of the wire between the poles is 3.45 cm.



When a direct current of 3.2 A is passed through the rod from P to Q , the reading on the digital balance increases by 2.69 grams.

- (a) Identify and explain the polarity of the magnet at pole X .

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- (b) Calculate the magnetic field strength between the poles of the magnet.

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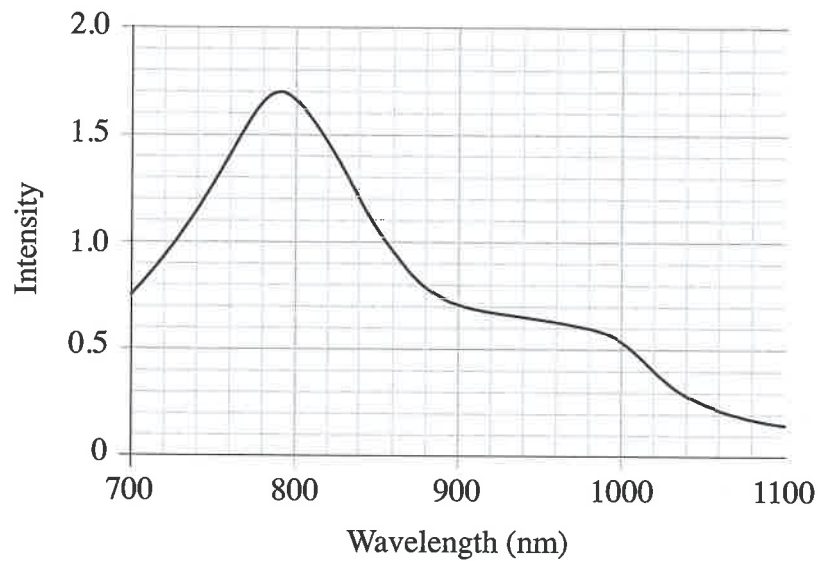
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Question 22 (4 marks)

30 Piscium is a red giant star in the zodiac constellation of Pisces that is visible to the naked eye. The spectrum below was recorded by the Cassini space probe in 2006.



- (a) Calculate the surface temperature of 30 Piscium.

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- (b) Explain how light from 30 Piscium can be used to provide evidence of its chemical composition.

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Question 23 (7 marks)

If hydrogen gas is bombarded by neutrons, a neutron can be captured, forming a deuterium atom ${}^2_1\text{H}$ with the emission of a gamma ray photon.

- (a) Write a nuclear equation for this reaction.

2

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- (b) Complete the table below showing the number of protons and neutrons in hydrogen and deuterium.

2

	Hydrogen ${}^1_1\text{H}$	Deuterium ${}^2_1\text{H}$
Proton number		
Neutron number		

- (c) By referring to the datasheet, calculate the rest mass of a neutron in atomic mass units (u).

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- (d) Use the data in the table below to calculate the energy carried by the gamma ray photon in MeV.

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Particle	Rest mass (u)
Hydrogen	1.0079
Deuterium	2.0141

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Question 24 (7 marks)

In December 2018, the comet 46P/Wirtanen passed the Earth while orbiting the Sun at its closest distance of 11.6 million km from the centre of the Earth. The comet's mass was estimated to be 2.1×10^{11} kg and it passed the Earth at this distance with a velocity of about 10 km s^{-1} .

- (a) Calculate the force between the comet and Earth at this distance.

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- (b) Calculate the theoretical orbital velocity at this distance required for a circular orbit around the Earth.

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- (c) Calculate the escape velocity from the Earth for the comet at this distance and use it to explain why the comet passes the Earth without being drawn into the Earth's orbit.

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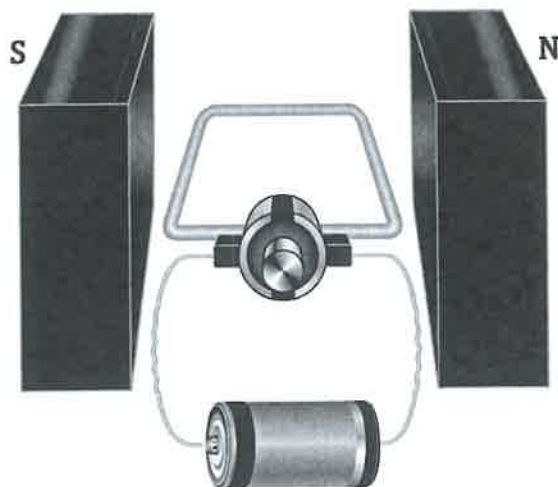
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Question 25 (7 marks)

A student constructed a simple DC motor as shown in the simplified diagram below. The plane of the rotating coil has an area of 0.0036 m^2 and the armature has 80 turns. The coil is placed in a uniform magnetic field of flux density 0.41 T and when a supply current of 1.5 A flows, the armature begins to turn.



- (a) Calculate the theoretical maximum torque produced by this motor.

1

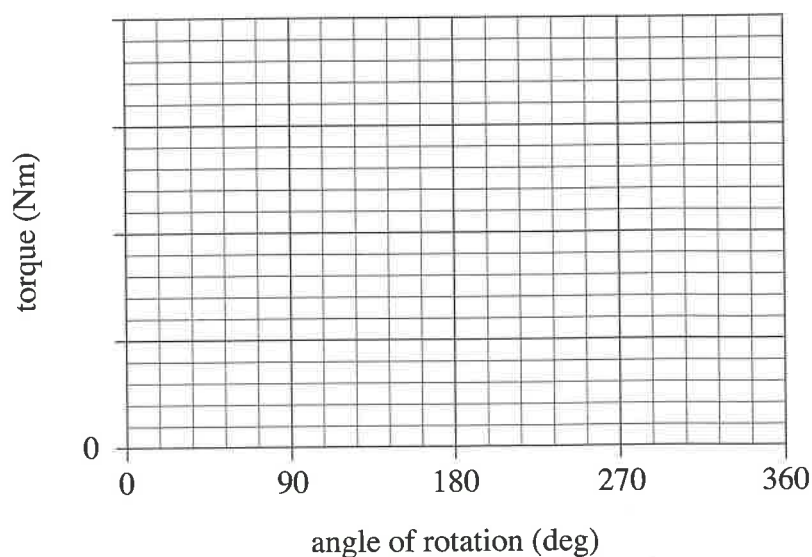
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- (b) Construct a graph to show how the torque varies through one revolution of the armature, from the starting position shown in the diagram.

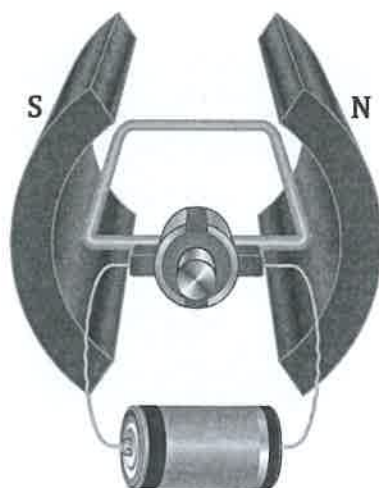
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Question 25 continued

- (c) To improve the torque, the student varied the construction of the motor by using radial magnets as shown.

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Use the grid in part (b) to show how this variation affects the torque and justify your answer below.

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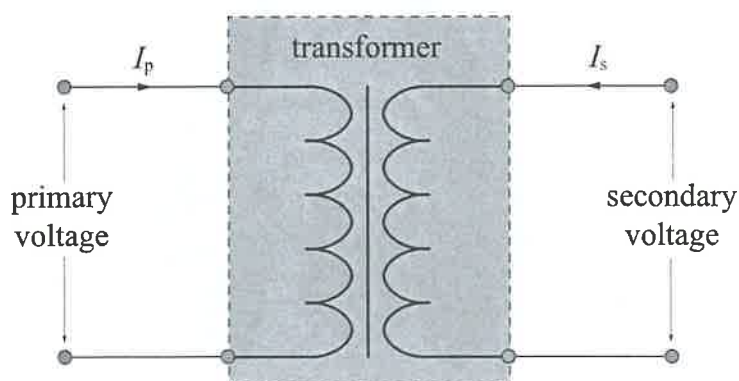
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Question 26 (4 marks)

In the electricity grid, a transformer is used to reduce voltage to 240 V for household use. The specifications of the transformer are shown below.

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In routine testing, a technician measures the input and output voltage and current for the transformer over a number of trials. The average results are shown in the table below.

	<i>average voltage (V)</i>	<i>average current (A)</i>
primary coil	7540	0.35
secondary coil	240	9.98

Account for these results in terms of the law of conservation of energy.

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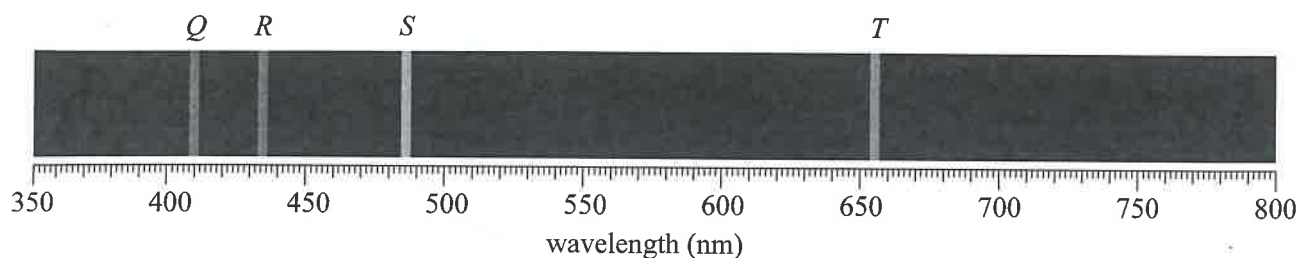
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Question 27 (7 marks)

A student researching the spectra of the various elements found the image below showing the Balmer series in a spectrum of hydrogen.



- (a) Outline how this spectrum might have been obtained experimentally.

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- (b) Interpret and compare the lines *Q*, *R*, *S* and *T* in terms of electron transitions and energy.

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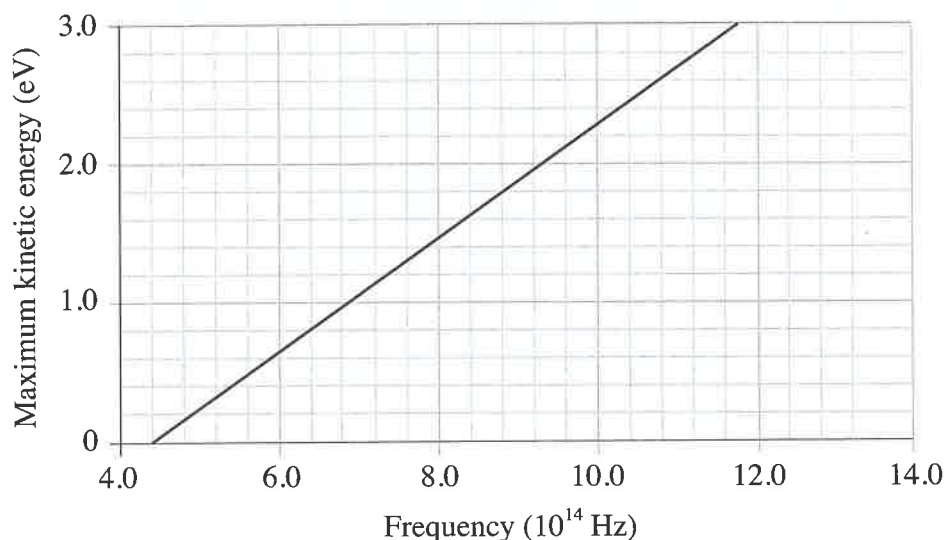
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Question 28 (7 marks)

In early experiments to investigate the photoelectric effect, a beam of monochromatic light was directed at a clean surface of potassium metal. By applying a stopping voltage to reduce the photocurrent to zero, the maximum kinetic energy of the ejected electrons was measured.

When the experiment was repeated with different frequencies of light, the maximum kinetic energy of electrons depended on the frequency of the light as shown below.



- (a) Calculate the wavelength of light required to begin photoemission.

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- (b) In another experiment, the potassium metal surface was replaced with aluminium metal with a work function of 4.2 eV.

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Calculate the threshold frequency for the aluminium surface.

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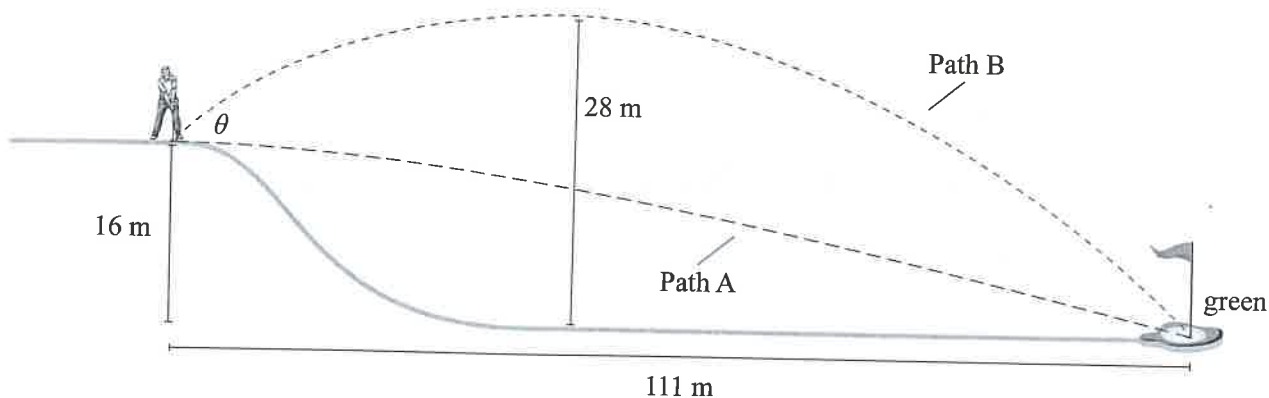
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- (c) On the graph above, draw a line to show the relationship between the maximum kinetic energy of ejected electrons and the frequency of incident light for the aluminium surface.

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Question 29 (6 marks)

A golfer launches a golf ball at a height of 16 metres vertically above the green which is 111 metres away. When launched horizontally, the golf ball follows Path A and when launched at an angle θ to the horizontal, it follows Path B.



- (a) Calculate the initial velocity of the golf ball when launched horizontally following Path A. 2

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- (b) Calculate the total time of flight when the golf ball is launched at an angle to the horizontal following Path B. 2

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- (c) Calculate the launch angle θ required for the ball to follow Path B. 2

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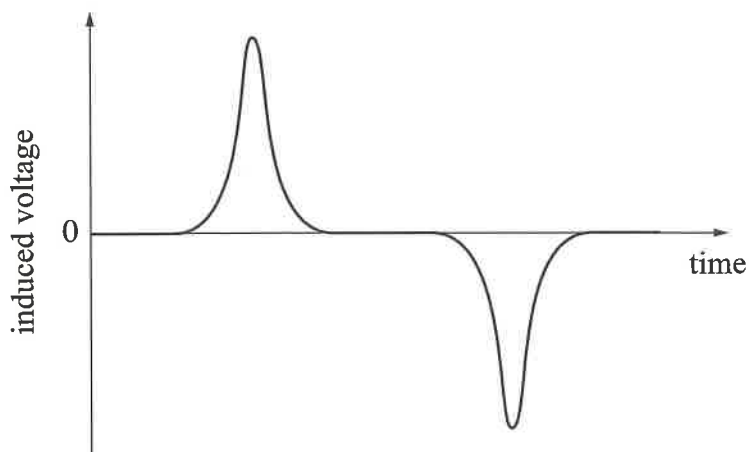
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Question 30 (5 marks)

A student conducted an investigation to verify Faraday's law of electromagnetic induction and presented some of the results in the graph shown below.

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Identify equipment, and describe and justify a procedure that would produce these results.

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Question 31 (7 marks)

An observer on Earth sees a spacecraft moving across the sky parallel to the horizon with a uniform speed of $2.3 \times 10^8 \text{ m s}^{-1}$. The spacecraft fires a missile of rest length 1.8 metres in the direction of its motion which is observed from Earth to be travelling across the sky with speed of $2.4 \times 10^8 \text{ m s}^{-1}$.

- (a) Calculate the length of the missile after it leaves the spacecraft as measured by the observer on Earth. 2

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- (b) The spacecraft now fires a laser in the direction of the missile. 5

A physics student assumes that an observer on Earth would measure the speed of the photons in the laser as the sum of the speed of light and the speed of the spacecraft.

Evaluate this claim.

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Question 32 (8 marks)

A car rounds a bend on a road that follows the arc of a circle with radius 25.6 metres. The car has a mass of 885 kg and the force of kinetic friction between the tyres and the road is 9420 N.

- (a) At what maximum speed should the car be moving so as to stay on the road while taking the bend? 1

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- (b) Using physics principles and your answer to part (a), quantitatively explain why motorists are advised to drive at slower speeds during wet weather. 3

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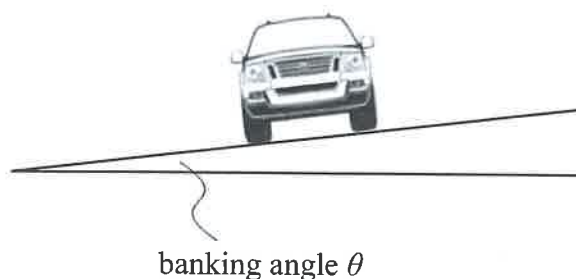
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Question 32 continued

- (c) For safety reasons, when building roads, corners are sometimes banked so they form an angle to the horizontal as shown in the diagram below. 4



If the bend in part (a) were banked at 8.0° to the horizontal, show quantitatively how this increases the maximum speed that allows the car to remain on the road.

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Jonathan Saurine (Convenor)
Lara Tyler

EXAMINERS

All Saints Grammar School, Belmore
All Saints Grammar School, Belmore

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Physics

DATA SHEET

Charge on electron, q_e	$-1.602 \times 10^{-19} \text{ C}$
Mass of electron, m_e	$9.109 \times 10^{-31} \text{ kg}$
Mass of neutron, m_n	$1.675 \times 10^{-27} \text{ kg}$
Mass of proton, m_p	$1.673 \times 10^{-27} \text{ kg}$
Speed of sound in air	340 m s^{-1}
Earth's gravitational acceleration, g	9.8 m s^{-2}
Speed of light, c	$3.00 \times 10^8 \text{ m s}^{-1}$
Electric permittivity constant, ϵ_0	$8.854 \times 10^{-12} \text{ A}^2 \text{ s}^4 \text{ kg}^{-1} \text{ m}^{-3}$
Magnetic permeability constant, μ_0	$4\pi \times 10^{-7} \text{ N A}^{-2}$
Universal gravitational constant, G	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Mass of Earth, M_E	$6.0 \times 10^{24} \text{ kg}$
Radius of Earth, r_E	$6.371 \times 10^6 \text{ m}$
Planck constant, h	$6.626 \times 10^{-34} \text{ J s}$
Rydberg constant, R (hydrogen)	$1.097 \times 10^7 \text{ m}^{-1}$
Atomic mass unit, u	$1.661 \times 10^{-27} \text{ kg}$ $931.5 \text{ MeV}/c^2$
1 eV	$1.602 \times 10^{-19} \text{ J}$
Density of water, ρ	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$
Wien's displacement constant, b	$2.898 \times 10^{-3} \text{ m K}$

FORMULAE SHEET

Motion, forces and gravity

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

$$\Delta U = mg\Delta h$$

$$P = \frac{\Delta E}{\Delta t}$$

$$\sum \frac{1}{2}mv_{\text{before}}^2 = \sum \frac{1}{2}mv_{\text{after}}^2$$

$$\Delta \vec{p} = \vec{F}_{\text{net}} \Delta t$$

$$\omega = \frac{\Delta \theta}{t}$$

$$\tau = r_{\perp} F = rF \sin \theta$$

$$v = \frac{2\pi r}{T}$$

$$U = -\frac{GMm}{r}$$

$$v = u + at$$

$$\vec{F}_{\text{net}} = m\vec{a}$$

$$W = F_{\parallel} s = Fs \cos \theta$$

$$K = \frac{1}{2}mv^2$$

$$P = F_{\parallel} v = Fv \cos \theta$$

$$\sum m\vec{v}_{\text{before}} = \sum m\vec{v}_{\text{after}}$$

$$a_c = \frac{v^2}{r}$$

$$F_c = \frac{mv^2}{r}$$

$$F = \frac{GMm}{r^2}$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

Waves and thermodynamics

$$v = f\lambda$$

$$f = \frac{1}{T}$$

$$d \sin \theta = m\lambda$$

$$n_x = \frac{c}{v_x}$$

$$I = I_{\text{max}} \cos^2 \theta$$

$$Q = mc\Delta T$$

$$f_{\text{beat}} = |f_2 - f_1|$$

$$f' = f \frac{(v_{\text{wave}} + v_{\text{observer}})}{(v_{\text{wave}} - v_{\text{source}})}$$

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$\sin \theta_c = \frac{n_2}{n_1}$$

$$I_1 r_1^2 = I_2 r_2^2$$

$$\frac{Q}{t} = \frac{kA\Delta T}{d}$$

FORMULAE SHEET (continued)

Electricity and magnetism

$$E = \frac{V}{d}$$

$$V = \frac{\Delta U}{q}$$

$$W = qV$$

$$W = qEd$$

$$B = \frac{\mu_0 I}{2\pi r}$$

$$B = \frac{\mu_0 NI}{L}$$

$$\Phi = B_{\parallel} A = BA \cos \theta$$

$$\mathcal{E} = -N \frac{\Delta \Phi}{\Delta t}$$

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

$$\vec{F} = q\vec{E}$$

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

$$I = \frac{q}{t}$$

$$V = IR$$

$$P = VI$$

$$F = qv_{\perp} B = qvB \sin \theta$$

$$F = I l_{\perp} B = I l B \sin \theta$$

$$\frac{F}{l} = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{r}$$

$$\tau = n I A_{\perp} B = n I A B \sin \theta$$

$$V_p I_p = V_s I_s$$

Quantum, special relativity and nuclear

$$\lambda = \frac{h}{mv}$$

$$K_{\max} = hf - \phi$$

$$\lambda_{\max} = \frac{b}{T}$$

$$E = mc^2$$

$$E = hf$$

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$l = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

$$p_v = \frac{m_0 v}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$N_t = N_0 e^{-\lambda t}$$

$$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$$

PERIODIC TABLE OF THE ELEMENTS

1 H 1.008 Hydrogen		4 Be 9.012 Beryllium		79 Au 197.0 Gold		2 He 4.003 Helium	
3 Li 6.941 Lithium		12 Mg 24.31 Magnesium		Atomic Number Symbol Name		5 B 10.81 Boron	
11 Na 22.99 Sodium		20 Ca 40.08 Calcium		Standard Atomic Weight		6 C 12.01 Carbon	
19 K 39.10 Potassium		38 Sr 87.61 Strontium		Name		7 N 14.01 Nitrogen	
37 Rb 85.47 Rubidium		56 Ba 137.3 Barium		79 Au 197.0 Gold		8 O 16.00 Oxygen	
55 Cs 132.9 Cesium		88 Ra Radium		26 Fe 55.85 Iron		9 F 19.00 Fluorine	
87 Fr Francium		Actinoids		25 Mn 54.94 Manganese		10 Ne 20.18 Neon	
		21 Sc 44.96 Scandium		24 Cr 52.00 Chromium		17 Cl 35.45 Chlorine	
		39 Y 88.91 Yttrium		23 V 50.94 Vanadium		35 Br 79.90 Bromine	
		71 La 138.9 Lanthanum		22 Ti 47.87 Titanium		53 I 126.9 Iodine	
		89-103		40 Zr 91.22 Zirconium		84 Po Rn Radon	
		Actinoids		41 Nb 92.91 Niobium		85 At Ts Tennessine	
		Rutherfordium		42 Mo 95.96 Molybdenum		Livermorium	
		Dubnium		43 Tc Technetium		Flerovium	
		Seaborgium		44 Ru 101.1 Ruthenium		Oganesson	
		Bohrium		45 Rh 102.9 Rhodium			
		Hassium		46 Pd 106.4 Palladium			
		Meitnerium		47 Ag 107.9 Silver			
		Darmstadtium		48 Cd 112.4 Cadmium			
		Roentgenium		49 In 114.8 Indium			
		Copernicium		50 Sn 118.7 Tin			
				51 Sb 121.8 Antimony			
				52 Te 127.6 Tellurium			
				53 I 126.9 Iodine			
				54 Xe 131.3 Xenon			
				55 Cs 132.9 Cesium			
				56 Ba 137.3 Barium			
				57-71 Lanthanoids			
				72 Hf 178.5 Hafnium			
				73 Ta 180.9 Tantalum			
				74 W 183.9 Tungsten			
				75 Re 186.2 Rhenium			
				76 Os 190.2 Osmium			
				77 Ir 192.2 Iridium			
				78 Pt 195.1 Platinum			
				79 Au 197.0 Gold			
				80 Hg 200.6 Mercury			
				81 Tl 204.4 Thallium			
				82 Pb 207.2 Lead			
				83 Bi 209.0 Bismuth			
				84 Po Rn Radon			
				85 At Ts Tennessine			
				86 Rn Radon			
				87 Fr Francium			
				88 Ra Radium			
				89 Ac Actinium			
				90 Th 232.0 Thorium			
				91 Pa 231.0 Protactinium			
				92 U 238.0 Uranium			
				93 Np Neptunium			
				94 Pu Plutonium			
				95 Am Americium			
				96 Cm Curium			
				97 Bk Berkelium			
				98 Cf Californium			
				99 Es Einsteinium			
				100 Fm Fermium			
				101 Md Mendelevium			
				102 No Nobelium			
				103 Lr Lawrencium			

Lanthanoids

57 La 138.9 Lanthanum	58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.1 Ytterbium	71 Lu 175.0 Lutetium
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Actinoids

89 Ac Actinium	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium
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Standard atomic weights are abridged to four significant figures.

Elements with no reported values in the table have no stable nuclides.

Information on elements with atomic numbers 113 and above is sourced from the International Union of Pure and Applied Chemistry Periodic Table of the Elements (November 2016 version).

The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of all other data. Some data may have been modified.